

Chinese Activity Constructions: Light Verb Syntax and Adnominalisation:

Abstract:

In Chinese Activity constructions, the Activity nominal (Actor NOMINALIZER Activity) can appear either as the subject or the object of the Activity verb, in the latter case the subject usually consists of the Actor (Actor V Actor NOMINALIZER Activity). While the latter occurs in all Chinese dialects, the former is only found in northern dialects that generalise d-forms of the nominalizer, as southern dialects that use g-forms of the nominalizer prohibit the Activity nominal in subject position. In line with Huang's (2008) influential analysis in which subject Activity nominals are raised from object position, Tang (2008, 2009, 2011) proposes that *ge* blocks verb movement from the embedded clause in the Activity nominal to the matrix verb position, which is supposed to derive the ungrammaticality of *ge* in Activity nominals in subject position. Based on the technical difficulties of Huang and Tang's analyses, this paper proposes a new formal analysis in which Actor and Activity are merged in two separate positions in accordance with thematic principles (Ramchand (2008)), and the microvariation between *de* and *ge* is argued to be due to different nominalization strategies in which the Activity nominal contains either a nominal or clausal complement. The microvariation between *de* and *ge* can hence be derived from different selectional properties, and empirical support is offered from the history of Chinese as *de* and *ge* have different etymologies.

Keywords: Chinese, Mandarin, Cantonese, Activity, Nominalization

Introduction:

Mandarin Chinese has a nominal element *de* (的) known as adnominaliser or structural particle (結構助詞) which is commonly used to express possession and modification in the noun phrase (Paul (2005, 2012, 2019)), and it has been argued that *de* lexicalizes different functional heads in the nominal domain where possession (POSS) and modification (Mod) are hierarchically arranged along with quantificational and indexical categories such as demonstratives, numerals, quantifiers and classifiers (Aoun and Li (2003:140), Del Gobbo (2003:48), Cheng and Sybesma (2015:1520)):

1) Possessive Demonstrative Number Quantifier Classifier Modifier Noun

Furthermore, Mandarin Chinese regularly employs the adnominaliser *de* to construct nominal phrases in Activity constructions where the subject of the Activity verb is the Actor and the object is a nominal phrase headed by *de* containing the Actor and the Activity in a pseudo-possessive construction (Actor *de* Activity < Possessor *de* Possessum):

- 2) 他 當 他 的 老師
 ta dang ta de laoshi
 he do he DE teacher

'He is being a teacher (literally 'he is playing his role as a teacher').'

(Huang (2008:226), Tang (2011:149))

As the object (他的老師 *ta de laoshi* in 2)) denotes abstract activity (here 'him being a teacher') rather than possession ('his teacher'), the lexical noun (老師 *laoshi* in 2)) does not refer to a specific person or thing but an abstract activity, which has led some to argue that the Activity phrase here is a case of clausal nominalisation in Chinese where a clause denoting Activity is nominalised by *de* (Huang (2008:225), Hu (2016:435)). Moreover, it is generally possible to use the nominal phrase headed by *de* as the subject of Activity verbs, which, in the case of 2), can be rephrased as follows:

- 3) 他 的 老師 當 得 好
 ta de laoshi dang de hao
 he DE teacher do COMP good

'He is being a good teacher (literally 'he plays his role as a teacher well.')

(Huang (2008:225)) While in Mandarin Chinese the nominal phrase headed by *de* can occur in either subject (3) or object position (2) of the Activity verb, southern dialects of Chinese that use g-forms of the adnominaliser do not permit the Activity nominal headed by *ge* to occur in subject position (4b), even though the same

phrase can occur in object position (4a), which points to an important difference between the dialectal forms of the Chinese adnominaliser (Tang (2008, 2009, 2010, 2011)):

4a) 佢 做 佢 嘅 老師
 keoi zou keoi ge lousi
 he do he GE teacher

‘He is being a teacher.’

4b) *佢 嘅 老師 做 得 好
 keoi ge lousi zou dak hou
 he GE teacher be COMP good

‘He is being a good teacher.’ (Cantonese) (Tang (2008:75, 2009: 241-242, 2011:150))

This microvariation between adnominalisers *de* and *ge* points to a genuine difference between northern and southern dialects of Chinese, since Activity phrases nominalised by *ge* in southern dialects generally do not occur as the subject of Activity verbs (5d-f), even though they are widely attested in the object position (5a-c), as confirmed in Chinese dialectal surveys (Tang (2009)):

5a) 渠 當 渠 個 老師 (Shandong dialect (Xiang))
 keoi dang keoi ge laoshi
 he do he GE teacher

5b) 伊 當 伊 個 老師 (Taiwanese Min)
 yi dang yi ge laoshi
 he do he GE teacher

5c) 渠 當 渠 個 先生 (Wenzhou dialect (Southern Wu))
 keoi dang keoi ge xiansheng
 he do he GE teacher

‘He is being a teacher.’ (adapted from Tang (2009:242))

5d) *渠 個 老師 當 得 好 (Shandong dialect (Xiang))
 keoi ge laoshi dang de hao
 he GE teacher do COMP good

5e) *伊 個 老師 當 甲 好 (Taiwanese Min)
 yi ge laoshi dang ga hao
 he GE teacher do COMP good

5f) *渠 個 先生 當 好 險 (Wenzhou dialect (Southern Wu))
 keoi ge xiansheng dang hao xian
 he GE teacher do good PART

‘He is being a good teacher.’ (Tang (2009:241))¹

In sum, while it is acceptable in all Chinese dialects to use adnominalisers *de* and *ge* to nominalise Activity in the object position of Activity verbs (2, 4a, 5a-c), *de* but not *ge* can also head the Activity phrase in the subject position of the same Activity verbs (3, 4b, 5d-f). Adnominaliser *ge* seems more

¹ There are alternative structures in these dialects where subject Activity phrases can be headed not by the adnominaliser *ge* but by functors derived from lexical nouns that denote the instrument with which the Activity is carried out e.g. Cantonese:

1) 佢 手 籃球 打 得 好
 keoi sau laamkau da dak hou
 he hand basketball play COMP good
 ‘He plays basketball well.’

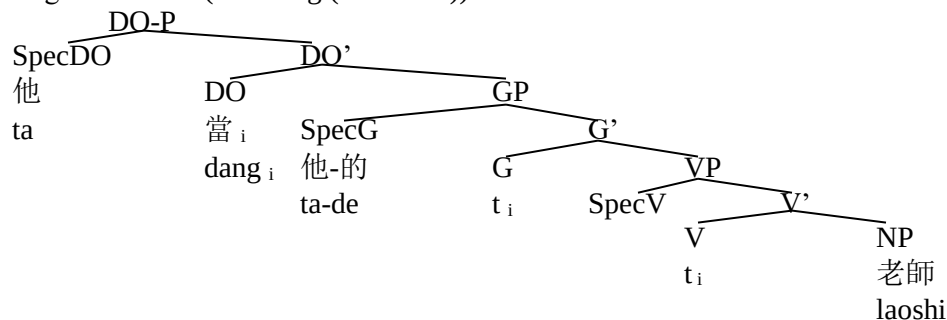
2) 佢 腳 波 踢 得 好
 keoi geot bo tek dak hou
 he foot ball kick COMP good
 ‘He plays football well.’ (Tang (2009:242))

marked than *de* as the former can only nominalise the Activity phrase in the object position of Activity verbs (4a-b, 5a-f) while the latter can do so in both the object and subject positions (2-3). The two structural minimal pairs here are hence the subject and object asymmetry in using *ge* as the nominaliser of Activity phrases (4a-b, 5a-f) and the contrast between *de* and *ge* as nominalisers of the Activity phrase in subject position (3, 4b), both of which have implications for the Chinese nominal domain (1).

Section 1: Chinese Activity Constructions (Huang (2008)) and Nominalization Parameter (Tang (2008, 2009)):

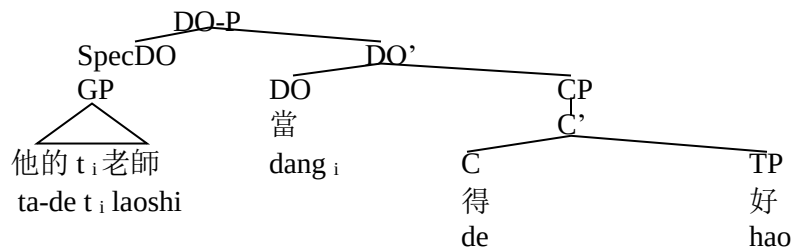
In order to account for the microvariation between *de* and *ge*, Tang (2008, 2009, 2010, 2011) proposes a nominalisation parameter based on Huang's (2008) formal analysis of Chinese Activity Constructions, which proposes a Light Verb projection DO for Activity verbs following Hale and Keyser (1993, 2002)). Huang (2008:226-232) argues that Light Verb DO selects a gerund phrase (GP) headed by the adnominaliser *de* which nominalises Activity by selecting a verb phrase, and when the verb in GP undergoes head movement to DO via the head of GP (G), this derives the canonical word order where the Actor and Activity phrase co-occur as the subject and object of the Activity verb respectively:

- 6) 他 當 他 的 老師
 ta dang ta de laoshi
 he do he DE teacher
 'He is being a teacher.' (cf Huang (2008:230))



Moreover, Huang (2008:230-232) argues that when the matrix Actor argument (他 *ta* in 6)) is deleted, it is possible for the verbless Activity GP (他的 *t_i* 老師 *ta de t_i laoshi* in 6)) to undergo remnant movement to the specifier of DO, which derives the subject and object alternation of the Activity phrase. Furthermore, this leaves room for the Activity verb to select a different complement such as the adverbial phrase 得好 *dehao* 'good, well' (Huang (2008:230-232)):²

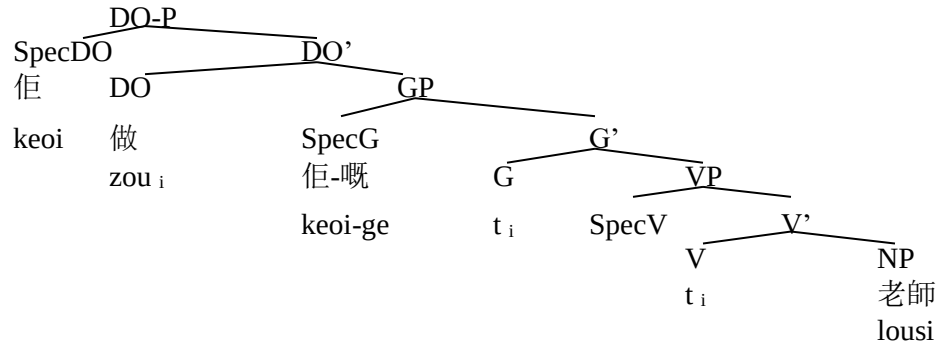
- 7) 他 的 老師 當 得 好
 ta de laoshi dang de hao
 he DE teacher do COMP good
 'He is being a good teacher.' (Huang (2008:231))



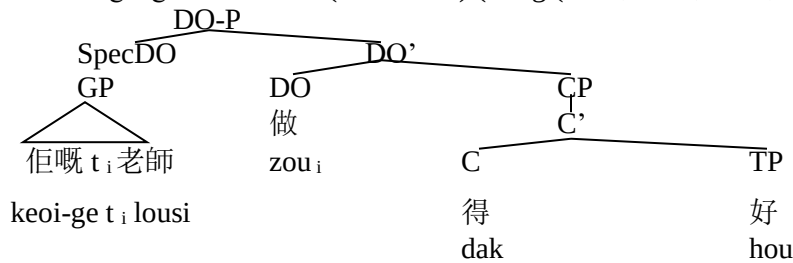
² Chinese regularly uses non-thematic complements to express adverbial notions such as duration, frequency and extent (Li (1990:7-10)) which, in accordance with Projection Principles such as Baker's UTAH, are merged below all thematic arguments of the verb (Lin (2001)).

Adopting Huang's derivational model, Tang (2008, 2009) proposes a nominalization parameter where the markedness of Cantonese *ge* is accounted for in terms of selectional properties, since it is argued that while Mandarin *de* can select clausal complements (8a) (Si (2002)), Cantonese *ge* cannot (8b), which is taken to support a nominalization parameter where *de* but not *ge* permits verb head movement from GP to DO via G, and this accounts for why *de* but not *ge* can nominalise Activity phrases in subject position (8c-d) (Tang (2008:76-77, 2009:243-244)):

- 8a) 張三 的 曾 在 美國 留學
 Zhangsan de ceng zai meiguo liuxue
 Zhangsan DE once at America study.abroad
 'Zhangsan's studying in the US in the past' (Si (2002:35))
- 8b) *張三 嘅 曾經 响 美國 留學
 Zeongsaam ge cangging heong meigwok lauhok
 Zeongsaam GE once at America study.abroad
 'Zeongsaam's studying in the US in the past' (Tang (2008:77))
- 8c) 佢 做 佢 嘅 老師
 keoi zou keoi ge lousi
 he be he GE teacher
 'He is being a teacher.'



- 8d) *佢 嘅 老師 做 得 好
 keoi ge lousi zou dak hou
 he GE teacher be COMP good
 'He is being a good teacher.' (Cantonese) (Tang (2008, 2009, 2010, 2011))



There are numerous problems in Huang and Tang's analyses. First of all, the categorial and syntactic status of adnominaliser *de* (6) and by extension *ge* (8c) is unclear, since Huang's (2008:230) analysis hinges on the hypothesis that *de* is cliticised onto the Actor argument in SpecGP (6) thus permitting head movement of the embedded verb to DO via G (當 *dang* _i ... *t* _i in 6), 做 *zou* _i ... *t* _i in 8c)). However, this clitic analysis is counterintuitive to the conventional analysis of *de/ge* as nominal(ising) heads of the Activity GP (1) (Tang (2006), Li (2008)), in which case *de/ge* (G) should block head movement of the embedded verb to DO by locality and minimality of movement (Rizzi (1990)). Secondly, the subject and object alternation of the Activity phrase (7, 8d) is not independently motivated, as no explanation is given for the deletion of the Actor argument in SpecDO-P and the subsequent remnant movement of the

Activity phrase, which seems to be an *ad-hoc* circular argument to account for the empirical data.³ Finally, and most critically, it is unclear how Tang's nominalization parameter can account for the minimal contrast between using *de* (7) and *ge* (8d) in the Activity phrase in subject position, since even if *de* and *ge* were both cliticised onto the Actor argument in SpecGP and hence permitted V-to-G-to-DO movement, it is unclear how and why *ge* but not *de* would block such head movement without positing extra and unnecessary stipulations. The dialectal microvariation between *de* and *ge* as seen in the subject-object asymmetry in the nominalisation of Chinese Activity phrases is a real empirical phenomenon that remains to be explained adequately, and the formal and empirical properties of Chinese adnominalisers (*de/ge*) and Activity constructions are presently explored.

This paper proposes a new formal analysis of Chinese Activity constructions which falls out from the subject and object asymmetry in the dialectal forms of the adnominaliser (*de/ge*), namely the fact that while both *de* and *ge* can occur in the object position of Activity verbs (2, 4a), only *de* (3) and not *ge* (4b) can occur in the subject position of such verbs, which lies at the heart of the microparametric variation in Chinese dialects. There are three sections in the main part of this paper: section 2 proposes modifications to Huang and Tang's analyses by eliminating the *ad-hoc* remnant movement of the Activity phrase to SpecDO-P which is not only problematic but also violates traditional Projection Principles that assume a one-to-one correspondence between thematic arguments and A(argument)-positions (Hu (2016:437), cf Chomsky (1981:36), Stowell (1981:112), see footnote 3). Rather, it is argued that Actor and Activity are merged in two distinct A-positions where Actor is merged higher than Activity in line with traditional thematic hierarchies (Baker (1985, 1988)). Since Actor is the external argument in Activity constructions (Hale and Keyser (1993:77-84, 2002:88-103)), Actor and Activity can be argued to be merged in the specifier positions of Ramchand's (2008) InitiateP and ProcessP respectively, which conforms to the event structure of Activity where SpecInitiateP and SpecProcessP hold the agent and the theme respectively (Ramchand (2008:52ff)). The canonical base structure of Chinese Activity constructions thus consists of Actor and Activity as two independent arguments, which provides a common derivational model (Initiate + Process) for examining the microparametric variation between adnominalisers *de* and *ge*. In light of this new event structure, the subject and object asymmetry between *de* and *ge* in heading the Activity phrase can be remodelled as such: in the case of the canonical structure where both Actor and Activity co-occur, both *de* and *ge* can head the Activity phrase (SpecProcessP) as the object of the Activity verb if there is an Actor argument (SpecInitiateP) functioning as the subject above it, and if there is no such Actor argument (SpecInitiateP), only *de* and not *ge* can head the Activity phrase (SpecProcessP) as the subject of the Activity verb. The crux of this analysis is that ***ge* can head the Activity phrase (SpecProcessP) iff the Actor argument is projected (SpecInitiateP)**, which indicates a fascinating inextricable link between the Actor argument (SpecInitiateP) and the Activity phrase (SpecProcessP). In this analysis, the projection of the Actor argument seems to impose selectional restrictions on the nominalising head of the Activity phrase (*de/*ge*), which suggests that the Actor argument may be base-generated in the Activity phrase (SpecProcessP) and raised to its surface matrix A-position (SpecInitiateP), as seen in the coreference between the subject of the Activity verb and the external argument of the Activity phrase (他_i 當他_i 的老師 *ta_i dang ta_i de laoshi* (2) / 佢_i 做佢_i 嘅老師 *keoi_i zou keoi_i ge lousi* 'he is being a teacher' (4a)). In the absence of the matrix Actor argument, on the other hand, there is a fundamental difference in the Activity phrase which can only be nominalised by *de* and not by *ge* (他的老師當得好 *ta de laoshi dang de hao* (3) / *佢嘅老師做得好 *keoi ge lousi zou dak hou* 'he is being a good teacher' (4b)). Section 3 examines the etymological origins of *de* and *ge*, and while *de* is widely taken to derive from phrase-final nominaliser *di* (底) in nominal linking contexts (Feng (1990),

³ Professor Jim Wood (p.c.) has suggested to me that this alternation may be on a par with passivisation where the subject argument is suppressed and the object argument is moved to the subject position for Case-assignment (Burzio (1986)). However, while the subject/object alternation of the Activity phrase is schematically similar to the Case-driven raising of the object in passivisation, there is not an independent principle such as Case to derive it (cf Chomsky (1981)), which leaves it as yet unexplained.

Jiang (1999), cf den Dikken and Singhapreecha (2004)), *ge* is derived from the general classifier *ge* (個/固) which may be incompatible with nouns denoting abstract Activity since classifiers typically have count, individualising and delimiting properties that naturally select concrete and countable nouns (Bisang and Li (2012), Zhang (2013)). The microvariation between *de* and *ge* in Activity phrases can hence be explained, since *de* is compatible with a wider range of nouns and is attested with abstract nouns denoting Activity whereas *ge* has selectional restrictions on the noun ([+count]) and is incompatible with abstract nouns. This supports the hypothesis that the Activity phrase has a different internal structure depending on the projection of Actor (SpecInitiateP), in the absence of which the Activity phrase contains a nominal complement that is subject to the selectional differences between *de* and *ge*, the latter of which is incompatible with lexical nouns denoting abstract Activity due to its de-classifier origins. Section 4 concludes by examining the biclausal structure of Chinese Activity constructions which involves two types of nominalization, and although the Activity phrase is thematically constant and is merged in the same A-position (SpecProcessP), its internal structure can vary and correlate with the matrix Actor argument and the internal structure of the nominalised Activity phrase indicates that while the Activity argument may be thematically constant in being merged in the same A-position (SpecProcessP), it can have a different internal structure in containing either a nominal or clausal complement, which differs from traditional distinctions between result and process nominalisations (Grimshaw (1990)).

Section 2: Chinese Activity constructions: Actor + Activity:

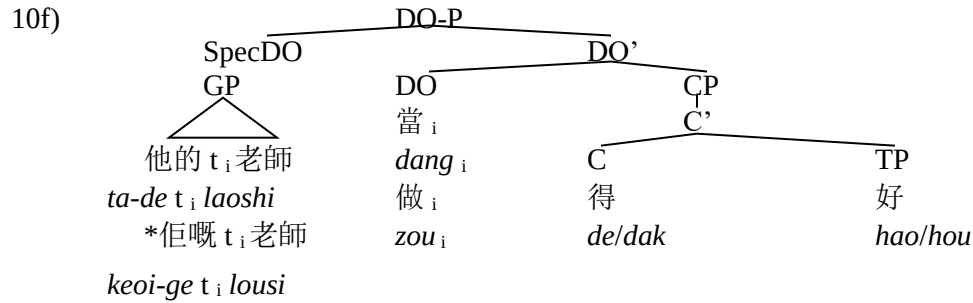
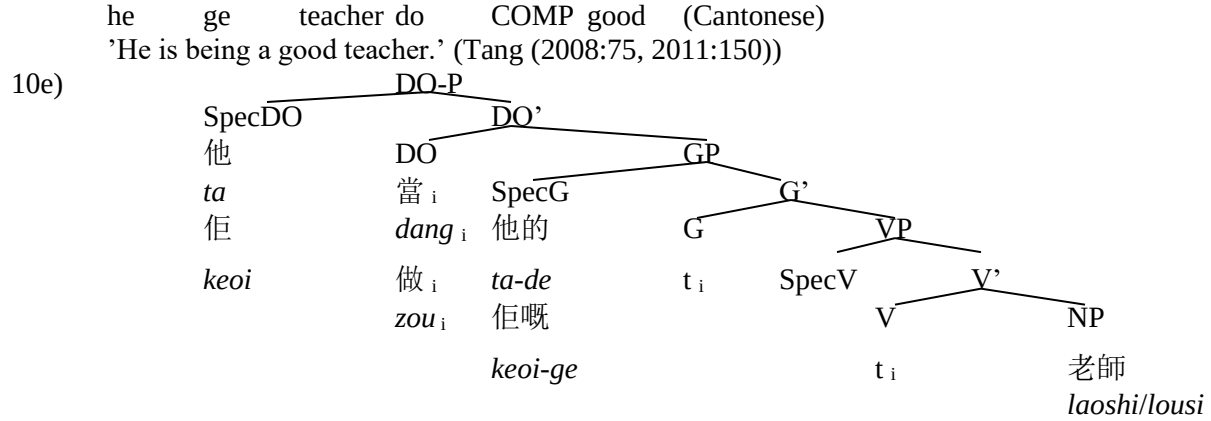
As shown in the introduction, all Chinese dialects have Activity constructions where the Actor and Activity occur in the subject and object positions of the Activity verb respectively (2, 4a), which seems to indicate two distinct A-positions for Actor and Activity, as repeated here:

- 9a) 他 當 他 的 老師
ta dang ta de laoshi
he do he DE teacher (Mandarin)
- 9b) 佢 做 佢 嘅 老師
keoi zou keoi ge lousi
he do he GE teacher (Cantonese)

‘He is being a teacher.’ (Huang (2008:226), Tang (2011:149))

While Huang (2008) seeks to unify Chinese Activity constructions with Activity constructions in other languages such as English and Japanese denominal constructions by postulating a Light Verb projection DO that introduces the Actor as its sole argument (SpecDO-P) and triggers head movement of the embedded verb in the GP complement, his analysis is problematic on two counts: 1) the postulation of a nominal Activity phrase (GP) as the complement to DO seems to be a categorial mistake as Light Verbs are merged above verbal projections and should thus select a verbal complement (Chomsky (1993), Hale and Keyser (1993, 2002), Cinque (1999)) 2) the subject and object alternation of the Activity phrase is not independently motivated as it is supposed to occur by deletion of the Actor argument in SpecDO-P and remnant movement of the Activity phrase (see footnote 3), which is shown here again (cf 6-8):

- 10a) 他 當 他 的 老師
ta dang ta de laoshi
he do he DE teacher (Mandarin)
- 10b) 佢 做 佢 嘅 老師
keoi zou keoi ge lousi
he do he GE teacher (Cantonese)
- ‘He is being a teacher.’ (Huang (2008:225-226), Tang (2011))
- 10c) 他 的 老師 當 得 好
ta de laoshi dang de hao
he DE teacher do COMP good (Mandarin)
- 10d) *佢 嘅 老師 做 得 好
keoi ge lousi zou dak hou

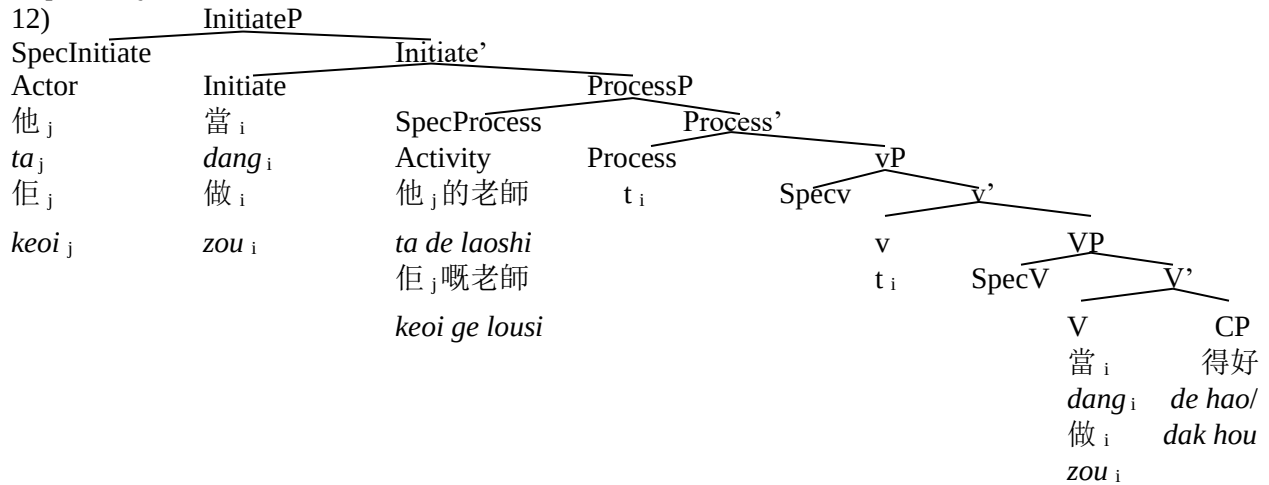


In 10f), Huang (2008:231-232) argues that the Actor argument (他 *ta*/佢 *keoi* in 10e)) is deleted and the verbless Activity GP undergoes remnant movement (他的 *t_i* 老師 *ta de t_i laoshi*/*佢嘅 *t_i* 老師 *keoi ge t_i lousi* in 10f)) to SpecDO-P, which makes it possible for Light Verb DO to select a different complement such as the adverbial complement phrase 得好 *de hao/dak hou* 'good, well' (see footnote 2). Furthermore, the raised Activity phrase can only be nominalised by *de* and not by *ge* (他的老師 *ta de laoshi*/*佢嘅老師 *keoi ge lousi* in 10f)), which Tang (2008 :76-77, 2009 :243-244) seeks to account for with his nominalization parameter that states that only *de* and not *ge* permits the embedded verb to move to DO via G (*t_i* in 10e)). It is unclear how Tang's nominalization parameter differentiates between *de* and *ge*, since even if these were cliticised onto the external argument in SpecGP and hence allowed V-to-G-to-DO movement, it is left unexplained how and why *de* and *ge* would differ in (dis)allowing this head movement. Furthermore, whether *de* or *ge* permits head movement of the embedded verb seems irrelevant to the subject and object positions of the Activity phrase, which are argued to be derived by remnant movement of the Activity phrase to SpecDO-P. Moreover, there is evidence that Actor and Activity may in fact be merged in two distinct A-positions, since Actor and Activity can co-occur with non-thematic complements such as adverbial clause 得好 *dehao* 'good, well', which renders the deletion of the Actor and remnant movement of the Activity phrase unnecessary as they could be base-generated in two different A-positions that dominate non-thematic complements (see footnote 2):

- 11a) 他 當 他 的 老師 當 得 好
ta dang ta de laoshi dang de hao
he do he DE teacher do COMP good (Mandarin)
- 11b) 佢 做 佢 嘅 老師 做 得 好
keoi zou keoi ge lousi zou dak hou
he do he GE teacher do COMP good (Cantonese)

‘He is being a good teacher.’⁴

It is hence possible to argue that Chinese Activity constructions consist of two separate and independent arguments (Actor, Activity) merged in two A-positions, which not only bypasses the minimality/locality issues in how the embedded verb can move to DO via G if G is lexically filled by *de/ge* but is also consistent with conventional Projection Principles whereby Actor is merged higher than Activity due to thematic prominence (Baker (1985, 1988), Dowty (1991)). Within a formal event structure where the agent who initiates the action is merged higher than the theme being affected, it can be argued that Actor and Activity are merged in the specifier positions of Ramchand’s (2008:39-42) InitiateP and ProcessP respectively, as shown below:

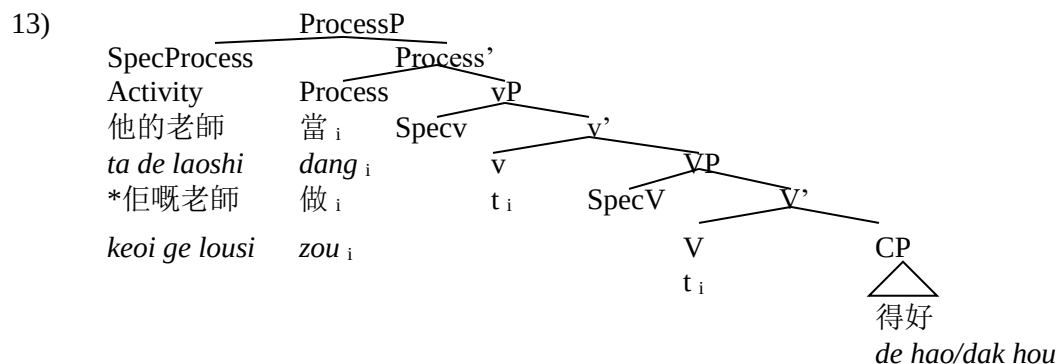


In 12), there are two Light Verb projections (Initiate, Process) that trigger verb movement (當 *dang*_i / 做 *zou*_i... *t*_i) and introduce two arguments which, in a bottom-up derivation, consist of Activity (SpecProcessP) and Actor (SpecInitiateP). In line with the Spec-Head A(rgument)-licensing of Light Verbs (Larson (1988)), it can be argued that Actor is copied and raised from the specifier of the Activity phrase below (他 *ta*_j / 佢 *keoi*_j), which entails that the Activity phrase contains an embedded clause in which Actor is base-generated and introduced by the same functional head (InitiateP). The derivation of the canonical word order of Chinese Activity constructions where both Actor and Activity are expressed as the subject and object of the Activity verb respectively is hence biclausal and the Activity phrase here is a case of clausal nominalisation where the adnominalizer (*de/ge*) selects a clausal complement. In the absence of Actor as seen in examples where the Activity phrase occurs as the subject of the Activity verb, Activity is the only argument (SpecProcessP), which similarly triggers verb head movement (V-to-Process), as shown below:

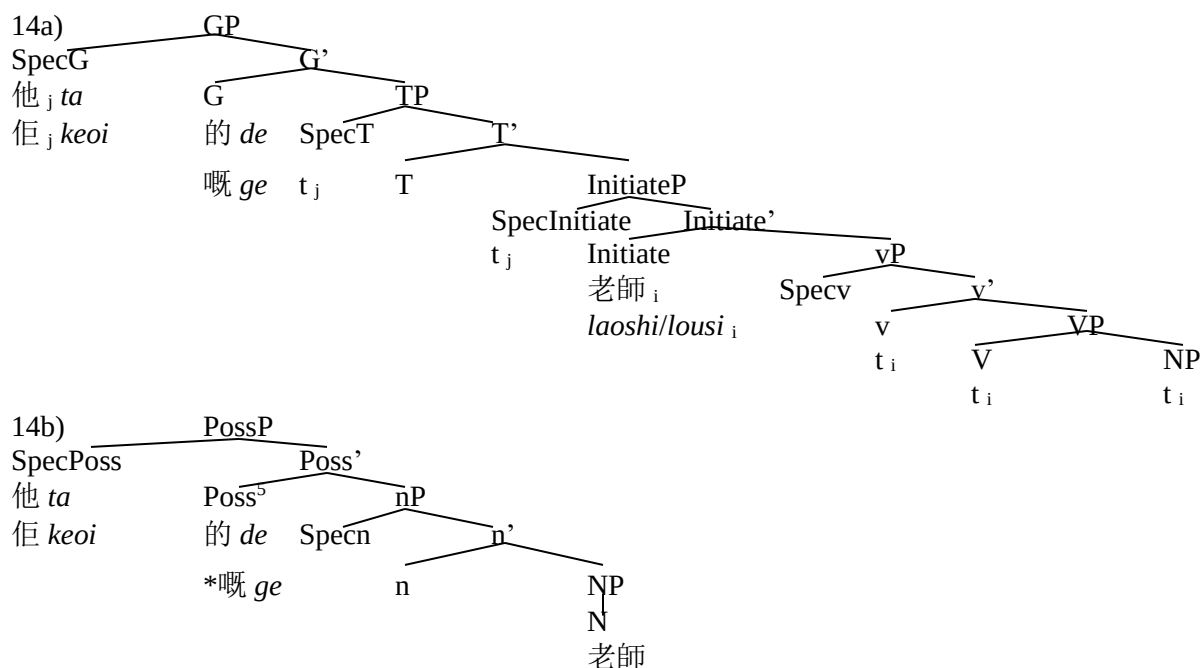
⁴ It has been remarked to me by other native speakers that examples like 11a-b) sound a little stilted and should be marked by ?. In fact, the grammaticality improves when the Activity phrase is simplified by omitting the possessor (= Actor) of the phrase:

- i) 他 當 老師 當 得 好
 ta dang laoshi dang de hao
 he do teacher do COMP good
- ii) 佢 做 老師 做 得 好
 keoi zou lousi zou dak hou
 he do teacher do COMP good
 ‘He is being a good teacher.’ (Mei (1978))

Nonetheless, it remains the case that it is possible to express Actor and Activity together with an adverbial complement, which indicates two distinct A-positions here.



In contrast to 12), only *de* and not *ge* can head the Activity phase (他的老師 *ta de laoshi*/*佢嘅老師 *keoi ge lousi* in 13)), which indicates an interesting connection between the matrix Actor argument (SpecInitiateP) and the Activity phrase (SpecProcessP), since in the absence of the former the latter can only be headed by *de* and not by *ge* (13). The internal structure of the Activity phrase seems sensitive not only to the selectional properties of the adnominaliser (*de*/**ge*) but also to the projection of the matrix Actor argument, which suggests that the Activity phrase in 13) is not clausal nominalization as in 12) but a nominal possessive phrase containing a lexical noun denoting Activity. The two structures of the Activity phrase are represented below:

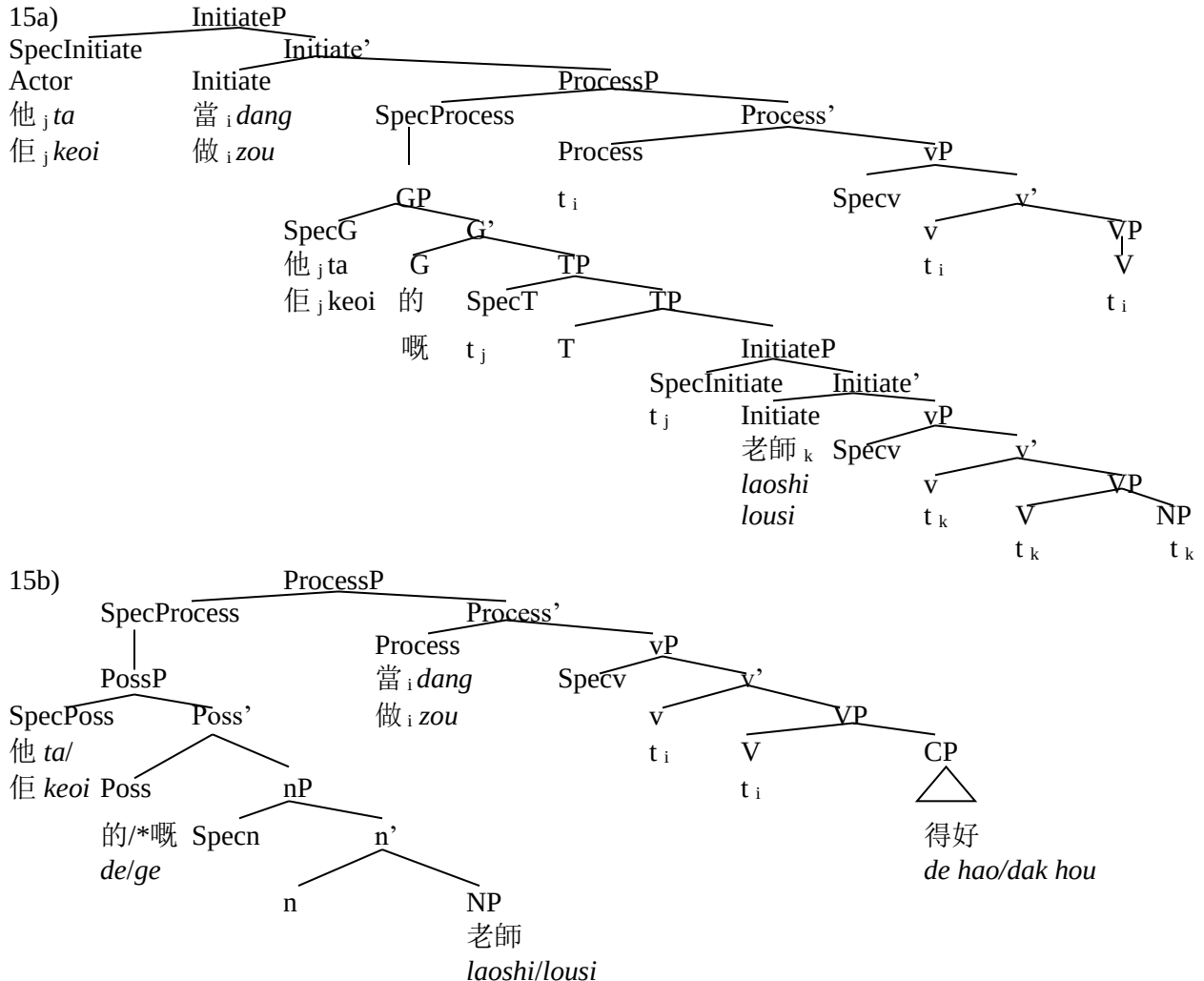


⁵ Huang's (2008) labels the nominal head of clausal nominalization as G(erund), which corresponds to the various D-projections in the nominal domain that can also introduce arguments in their specifiers. In main text, 14a), Actor is base-generated in the embedded SpecInitiateP and moves to SpecGP, while in main text, 14b), G corresponds to Poss which introduces the possessor to the noun phrase (Longobardi (2001), Longobardi and Silvestri (2013)).

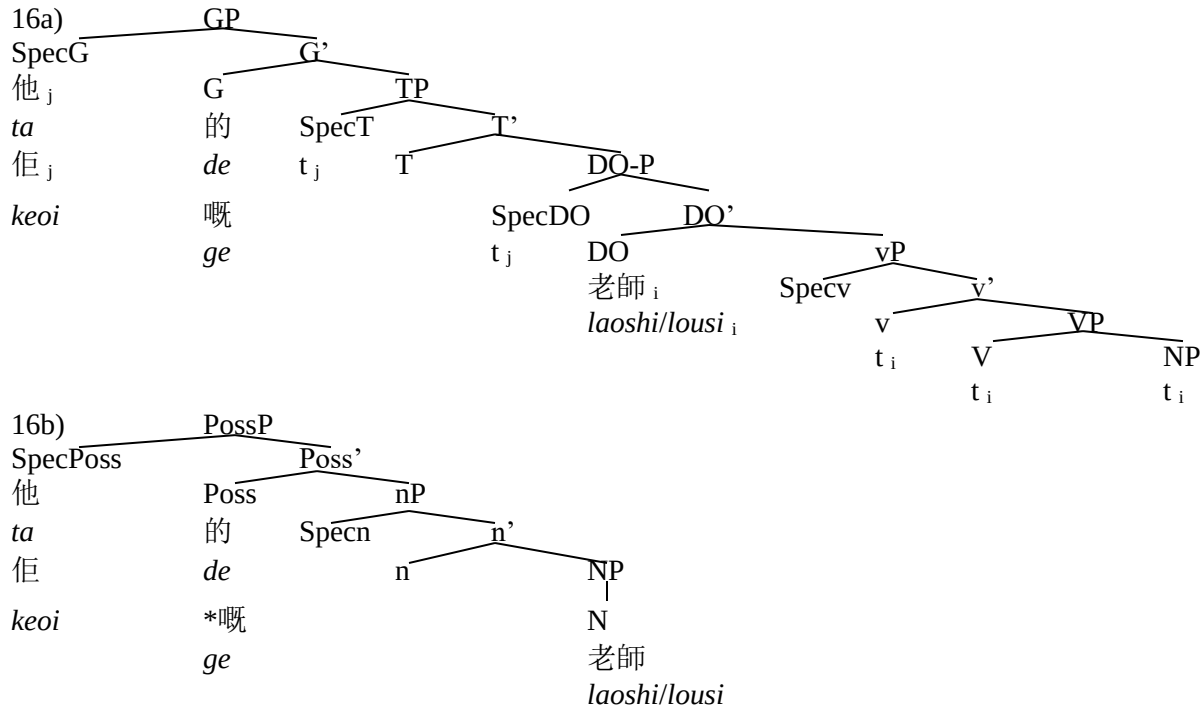
In sum, while both *de* and *ge* can nominalise a clause by selecting a clausal complement (14a), only *de* and not *ge* can select a nominal complement denoting Activity (14b), which shows that Chinese adnominalisers can have selectional restrictions on the lexical noun but not on the clausal complement (14a). These two types of Activity constructions hence consist of two different types of nominalization, namely clausal nominalization (14a) in the case where the matrix Actor argument is projected (12) and a nominal possessive phrase (14b) if the matrix Actor argument is not projected (13), which correlates with the microvariation between adnominalisers *de* and *ge* in heading the Activity phrase in subject position (13). This microvariation is examined in the next section where the etymology and selectional properties of *de* and *ge* are considered.

Section 3: Chinese adnominalisers *de* and *ge* : Activity Phrase:

The previous section proposed a biclausal structure for Chinese Activity constructions where the Activity phrase (SpecProcessP) is dominated by Actor (SpecInitiateP) (12), which is derived via head movement of the Activity verb (V-to-Process-to-Initiate), and Actor is argued to be copied and raised from the embedded clause in the Activity phrase (14a). When Actor is not projected (13), Activity becomes the sole argument in subject position (SpecProcessP) where it shows a different type of nominalization, namely a possessor noun phrase containing a nominal complement denoting Activity which can only be headed by *de* and not by *ge* (14b). The two types of Activity constructions are represented here again:



The two minimal pairs of interest are the subject and object asymmetry of the Activity phrase which can only be headed by *ge* (佢嘅老師 *keoi ge lousi*) in object (15a) but not in subject position (15b), and the Activity phrase in subject position (15b) which can only be headed by *de* (他的老師 *ta de laoshi*) and not by *ge* (*佢嘅老師 *keoi ge lousi*). The presence of the matrix Actor argument (SpecInitiateP) which seems to render possible the clausal nominalisation of *ge* is argued to indicate that the Activity phrase below (SpecProcessP) contains a clausal complement (TP) in which Actor is base-generated and raised to the matrix A-position (15a), whereas without the projection of Actor the Activity phrase contains not a clausal but a nominal complement (15b), which is shown here again (cf 14a-b):



The selectional differences between adnominalisers *de* and *ge* in selecting a noun denoting Activity (16b) can be derived from their etymological and historical origins. Adnominaliser *de* is historically derived from phrase-final nominaliser *di* (底) which has been argued to originate from Classical Chinese *zhe* (者) (Liu (2008), Aldridge (2009), Yap et al (2011)), and the various functions of *de* have been shown to originate from appositional constructions where phrase-final *di* is juxtaposed with a coreferential noun phrase which can be reanalysed as a single noun phrase where *di* is reanalysed as a phrase-medial nominal linker:

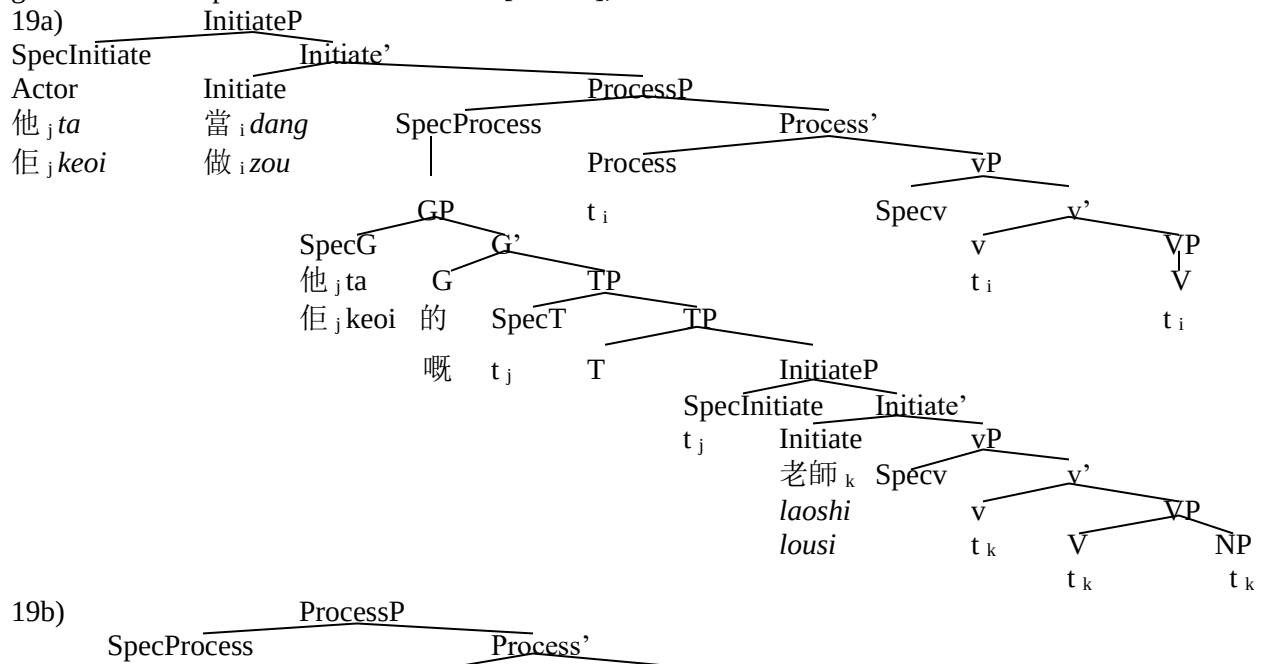
- 17) 昨 來 底 後生
 zuo lai di housheng
 yesterday come DI youth
 ‘the one who came yesterday, the youth’ > ‘the youth who came yesterday’ (祖堂集)

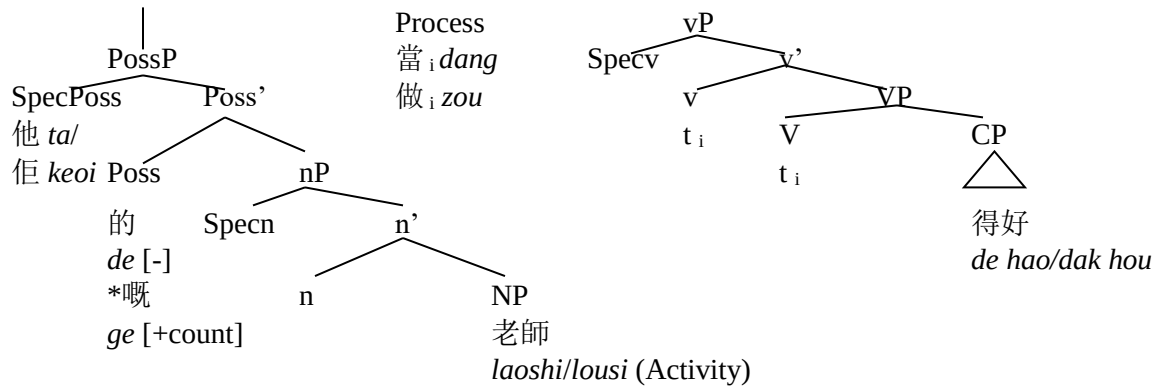
Adnominaliser *ge*, on the other hand, is derived from the general classifier *ge* (個/固), which is typically used for counting and quantification and hence inherently delimits its nominal complement (Bisang and Li (2012), Zhang (2013)). There are hence selectional differences between *de* and *ge*, and while *de* is attested with abstract nouns denoting skill/ability (18a-b), *ge* regularly selects concrete countable nouns (18c-e) (Cao (1995), Zhao (1999), Shi (2012), Li (2016)):

- 18a) 自己 的 鐮 與 槍 舞 得 好
 ziji de jian yu qiang wu de hao
 self DE arms and spear wield COMP good
 ‘He wields his arm and spears well’ (literally ‘his skills in using arms and spear are good’)

- (隋唐演義)
- 18b) 他 的 棋子 雖然 下 得 極 高
 ta de qizi suiran xia de ji gao
 he DE chess though play COMP very good
 'although he plays chess extremely well' (二十年目睹之怪現狀)
- 18c) 好 個 聰明 人
 hao ge congming ren
 good GE clever person
 'A good clever person' (敦煌變文集)
- 18d) 堂堂 好 個 丈夫-兒
 tangtang hao ge zhangfu-er
 indeed good GE man-DIM
 'good men indeed' (敦煌變文集)
- 18e) 後面 個 僧 祇 對 看
 houmian ge seng zhi dui kan
 behind GE monk only face look
 'the monk behind only looks in the face' (景德傳燈錄)

The collocation of *ge* with countable and delimited nouns (人 'person' (18c), 丈夫兒 'man' (18d), 僧 'monk' (18e)) and that of *de* with nouns that can be interpreted as abstract skill and activity (自己鏢與槍 'his wielding of arms and spears' (18a), 他的棋子 'his chess playing' (18b)) indicate different selectional properties that *de* and *ge* apply to their nominal complements, and while *de* can be used as the head of Activity phrases containing lexical nouns denoting abstract Activity (e.g. 鏢與槍 'arms and spear-wielding' (18a), 棋子 'chess-playing' (18b)), *ge* selects countable and delimited nouns which are incompatible with abstract Activity. These differences seem to have persisted into dialectal forms of the adnominaliser *de* and *ge* which show similar discrepancies in heading Activity phrases in subject position where the adnominaliser selects nominal complements (15b, 16b), and these selectional properties are non-applicable when *de* and *ge* head Activity phrases in object position where they contain an embedded clause (15a, 16a). The semantic markedness of *ge* can be traced to its classifier origins, which can be given a formal representation as a feature [+count], as shown below:





The selectional properties ([+count]) of the adnominalizer can account for the markedness of seem to have different on the lexical noun which affect their nominal selection of Activity. The next and final section makes concluding remarks on the syntax and semantics of the two types of Chinese Activity constructions and the internal structure of the Activity phrase.

Section 4: Chinese Activity Constructions: Light Verb Syntax and Nominalization:

This paper has proposed and defended a biclausal and bi-argumental analysis of Chinese Activity constructions based on the canonical structure where Actor and Activity are expressed as the subject and object of the Activity verb respectively, which is attested in all Chinese dialects, as shown below:

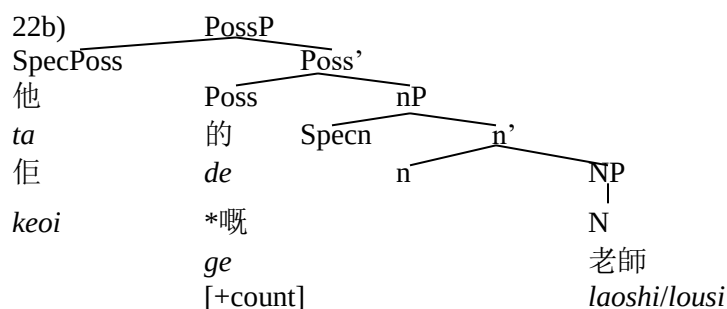
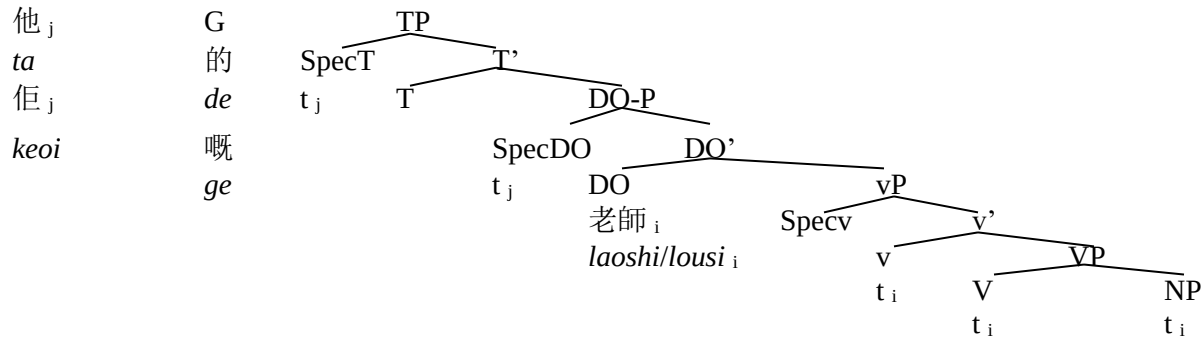
- 20a) 他 當 他 的 老師
ta dang ta de laoshi
he do he DE teacher (Mandarin)
- 20b) 佢 做 佢 嘅 老師
keoi zou keoi ge lousi
he do he GE teacher (Cantonese)
- 'He is being a teacher.' (Huang (2008:225-226), Tang (2011))

In these examples, the repetition of Actor (他_i 當 他_i 的老師 *ta_i dang ta_i de laoshi* in 20a), 佢_i 做 佢_i 嘅 老師 *keoi_i zou keoi_i ge lousi* in 20b)) is argued to be evidence for the base-generation of Actor in the Activity phrase (他的老師 *ta de laoshi* in 20a), 佢嘅老師 *keoi ge lousi* in 20b)), which thus constitutes clausal nominalization as the embedded clause has the same Light Verb projection (InitiateP) that mirrors the matrix clause (19a). The alternative Activity construction where there is no Actor but only Activity is mono-argumental, as shown below:

- 21a) 他 的 老師 當 得 好
ta de laoshi dang de hao
he DE teacher do COMP good (Mandarin)
- 21b) *佢 嘅 老師 做 得 好
keoi ge lousi zou dak hou
he ge teacher do COMP good (Cantonese)
- 'He is being a good teacher.' (Tang (2008:75, 2011:150))

As there is no Actor and hence no InitiateP, the Activity phrase does not contain an embedded clause but an abstract lexical noun, and the different selectional properties of the adnominalisers are such that only *de* and not *ge* can head the Activity phrase (他的老師 *ta de laoshi* in 21a), *佢嘅老師 *keoi ge lousi* in 21b)), as *ge* imposes delimitedness ([+count]) on its nominal complement which makes it incompatible with Activity (18c-e). The two types of Activity phrase are represented as follows (cf 16a-b):





The internal structure of the Activity phrase correlates with the Light Verb projections in the matrix clause, namely Initiate which introduces and projects the Actor argument, and the core of Chinese Activity constructions consists of Initiate triggering verb movement and the raising of Actor from the Activity phrase below, which conforms to well known examples of Activity constructions that are derived by head movement of the lexical verb or object noun to Light Verb DO that introduces Actor in its specifier (Hale and Keyser (1993, 2002)). Furthermore, the derivation of Activity nominals in Chinese (22a-b), whose internal structure is discernible not by derivational morphology but by the microvariation in adnominalisers *de* and *ge*, the former of which can head the Activity phrase in both subject (21a) and object (20a) positions whereas the latter can only head the Activity phrase in object (20b) and not in subject (21b) position, constitutes evidence for extra syntactic factors (e.g. the projection of InitiateP in the matrix clause) that affect nominalization, which is different from traditional postulation of different types of nominals (result/event) with unique internal structure (Grimshaw (1990:45ff)). More research is required to explore Chinese nominal(izations).

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